

Teorema 1. *Let p, M, c such that $\Delta - tp(c) = \{\phi \in \Delta : p \vdash \phi\}$. Then p prime.*

Proof. By contradiction, assume there exist $\phi_1, \dots, \phi_n \in P(\Delta)$ such that $p \vdash \bigvee_{i=1, \dots, n} \phi_i$ but for all $i = 1, \dots, n$ we have $p \not\vdash \phi_i$. Since $c \models p$ we must have $c \models \phi_i$ for some $i = 1, \dots, n$. Put ϕ_i in *DNF* to obtain $c \models \bigvee_{k=1, \dots, m} \bigwedge_{j=1, \dots, n_k} \psi_j^k$ where $\psi_j^k \in \Delta$ for all j, k . Let k such that $c \models \bigwedge_{j=1, \dots, n_k} \psi_j^k$. Then for all $j = 1, \dots, n_k$ we have $c \models \psi_j^k$. Then for all $j = 1, \dots, n_k$ we have $p \vdash \psi_j^k$ and consequently $p \vdash \phi_i$, contradiction. $\square$